

ViewDiff – 3D-Consistent Image Generation with Text-to-Image Models

1. Introduction & Motivation

Text-to-Image (T2I) diffusion models (e.g., Imagen, DALL·E 2) have shown excellent performance in generating high-quality 2D images from text prompts. However, **extending these models to 3D asset generation** poses significant challenges:

- Existing methods often produce **inconsistent views**, lacking geometric coherence across multiple perspectives.
- Many systems rely on **synthetic datasets**, which lack the realism of real-world imagery.
- Approaches like DreamFusion and ProlificDreamer require **optimization per object**, making them slow and limited.

Key Insight: Leverage large-scale pretrained T2I diffusion models as powerful 2D priors and fine-tune them on real-world **multi-view datasets** to enforce **3D-consistent generation** while retaining image quality and diversity.

2. Problem Definition

Objective: Generate a set of multi-view images that are:

- **3D-consistent** (same object across different views)
- **Photo-realistic** (matching real-world appearance)
- **Controllable** by either text input or reference images and camera poses

3. Method Overview

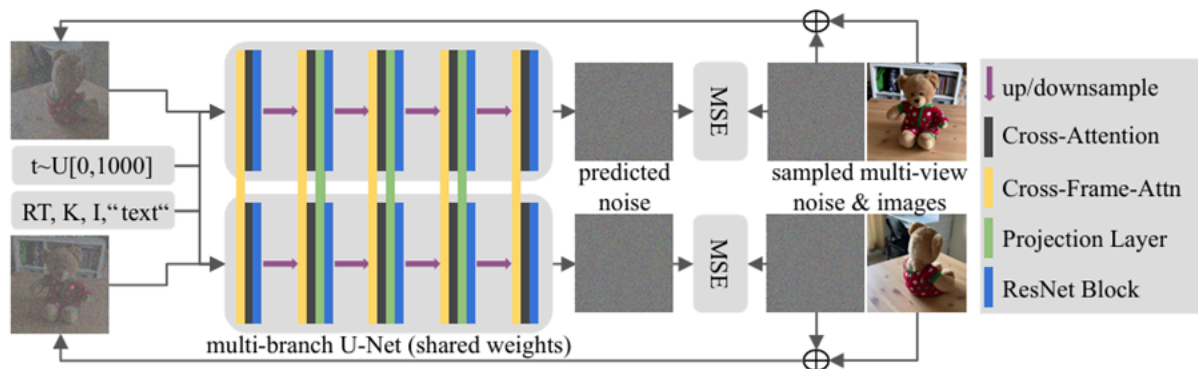
ViewDiff is built on pretrained 2D latent diffusion models and introduces two key architectural components:

1. **Cross-Frame Attention Layer (CFA)**
2. **Projection Layer with Volume Rendering**

These are embedded into each block of the U-Net architecture to learn **3D-aware image synthesis**.

Key Capabilities:

- **Autoregressive generation:** Generate additional views conditioned on previously generated ones.
- **Image or text conditioning:** Allows generation from either a single-view image or a descriptive prompt.



4. Architecture

a. Cross-Frame Attention (CFA)

- **Replaces standard self-attention** to enable multi-view understanding.
- **Matches spatial features** across all images in a batch to capture cross-view correlations.
- Incorporates:
 - **Camera pose**
 - **Intrinsic parameters**
 - **Image intensity encoding**
- **Purpose:**
Ensures that all generated views **share the same object identity**, maintaining **3D consistency** across frames.

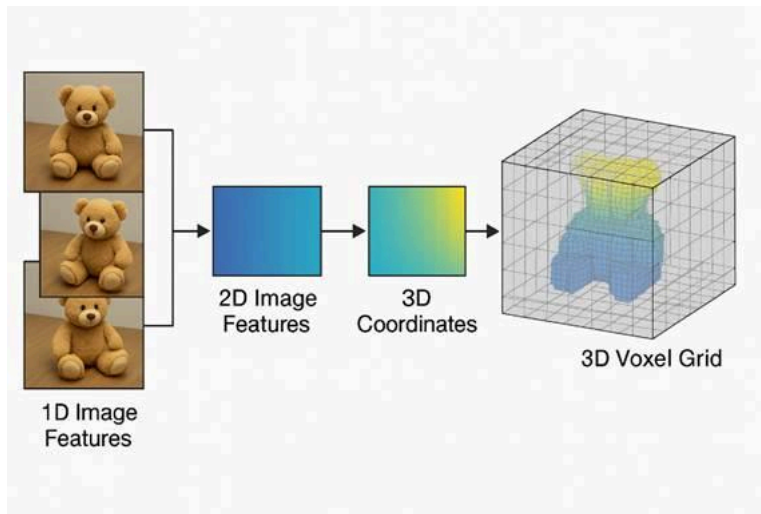
b. Projection Layer

- **Constructs a 3D voxel grid** (feature volume) from 2D feature maps.
- **NeRF-like volume rendering** to synthesize images from arbitrary viewpoints.

Include:

- **Foreground/background separation**
- **Raymarching** for accurate light accumulation

- Enables **precise viewpoint control** and ensures **3D-consistent feature rendering** across generated images.



5. Generation Modes

1. Unconditional Generation

- The **model automatically determines the timestep** for denoising.
- All samples start with **Gaussian noise**, and the model follows a predefined **timestep schedule** to refine them step by step.
- The **same timestep** is used across all samples in each iteration to maintain consistency.

2. Image-Conditional Generation

- The **user sets the timestep manually for the conditional part**.
 - Typically, the **given image's timestep is set to zero**.
- The **generative part starts with noise**, and the model follows a timestep schedule to refine it while conditioning on the input image.
- The model still **decides how to use the timesteps** for improving multi-view consistency.

So, in summary:

- For **unconditional generation**, the **model decides** the timestep.
- For **image-conditional generation**, the **conditional images have a fixed timestep (zero)**, but the **model still determines the rest dynamically**.

6. Training Strategy

- Trained on **CO3Dv2** dataset: real-world multi-view objects (e.g., hydrants, apples, donuts).

- Uses a mix of **prior preservation loss** (for maintaining 2D fidelity) and **denoising loss** (for 3D consistency).

7. Evaluation & Experiments

Trained Data:



Results:

These artifacts indicate that the model may not have seen enough high-quality examples of bear faces during training



This image illustrates how a model trained on insufficient or incomplete data can generate inconsistent or unrealistic outputs. Just as the bear's head and body don't quite match—creating an unnatural and disjointed appearance—models without access to comprehensive and

high-quality data often produce results that lack coherence or realism. The grainy texture further symbolizes the noise and artifacts that can emerge when data quality or quantity is lacking.

9. Challenges Faced During Implementation

1. **Hardware Limitations on Personal Laptops:**

The size and computational demands of the dataset exceeded the capabilities of standard personal laptops. The processing requirements, particularly for multi-view image generation and 3D rendering, necessitated high-performance GPUs such as the **NVIDIA RTX 3090** or equivalent. Without such hardware, training and inference were infeasible or extremely slow.

2. **Dependency and Framework Compatibility Issues**

Compatibility issues arose between various deep learning frameworks. Specifically:

- **TensorFlow** did not support several implementation techniques used in ViewDiff.
- Switching to **PyTorch and PyTorch3D** introduced its own set of version-related conflicts, requiring manual resolution of library mismatches and deprecated APIs.
- **RROR**: Failed building wheel for pytorch3d running setup.py clean for pytorch3d failed to build pytorch3d.
- Discarded existing setup.py created new setup.py creating and managing pytorch3d.

These inconsistencies disrupted the setup and increased the integration time.

3. **Data Preprocessing Bottlenecks Even on Smaller Datasets**

Despite opting for a significantly reduced dataset (~8.8 GB) instead of the full-scale CO3Dv2 dataset (~5.5 TB), the preprocessing and cleaning stages remained time-consuming. Initially, the smaller dataset was chosen to speed up experimentation. However, due to missing or inconsistent data in the reduced version that led to errors during model training, we had to proceed with the full 5.5 TB dataset to ensure data completeness and model reliability.

10. Limitations

- **Minor inconsistencies** in lighting/sharpness due to real-world dataset exposure variations.
- **Scene-scale generation** not addressed (only objects).
- **High memory usage** for autoregressive rendering of large view sets.

11. Conclusion

ViewDiff presents a significant leap in **photo-realistic and 3D-consistent image generation** by intelligently combining:

- The **expressivity** of 2D pretrained diffusion models.
- The **structure** of real-world 3D datasets.
- The **geometry-aware design** uses projection and cross-frame-attention layers.

It sets a new standard for **multi-view synthesis** from both images and text.